Justice-Chain SOS device

Components we have, which ones work, and what is still needed · 15 October 2026

We checked the nine photos of the parts we have against the earlier lists (*14-10-2026 Raspberry Pi shopping lists* and *14-10-2026 SOS device budget buying guide*). Each part below has its photo, so you can match it to the one in your hand.

> **The main board is a Raspberry Pi 4 Model B.** It is one of the supported boards, so we follow the **Pi 4 list** (Option 3 in the shopping lists, Option B in the buying guide). The code and wiring do not change: SOS button on pin 11, status LED on pin 13.

Verdict	Parts
Use it	1 Raspberry Pi 4 · 2 MB-102 breadboard
Use after checking	3 Resistors (measure first)
Not needed, keep for later	4 ADXL345 accelerometer
Not for this device	5 PT100 probe · 6 MAX31865 board · 7 SCT-013 current clamp · 8 Thermal paste · 9 5 V 1 A adapter

1. Parts we can use



1 Raspberry Pi 4 Model B (in the pink official case)

Photo: WhatsApp Image 2026-10-15 at 1.47.37 PM

Use it: this is the device

What it is: the small computer that runs the SOS program. The board says "Raspberry Pi 4 Model B ©2018". It sits in the base of the official pink-and-white case and already has three stick-on heatsinks.

- Has the **15-pin camera port** (marked CAMERA), the 40-pin GPIO header, 4 USB ports and Wi-Fi.
- Any RAM size works (1, 2, 4 or 8 GB). The size is printed on the memory chip or the box.
- Power goes in through the **USB-C** port (POWER IN) and needs **5 V 3 A**.
- The heatsinks and the case are done, so neither needs buying.



2 MB-102 solderless breadboard

Photo: WhatsApp Image 2026-10-15 at 1.46.49 PM

Use it

What it is: a full-size (830-hole) board for connecting the button, LED and resistor without soldering.

- Replaces the "mini breadboard" on our list. It is bigger than needed but works the same.
- Wires from the Pi need **female-to-male** jumpers (female end on the Pi pin, male end in the breadboard). We still need those.



3 Resistors (4 pieces, orange-orange-black bands)

Photo: WhatsApp Image 2026-10-15 at 1.49.11 PM

Use after checking the value

What it is: a resistor limits the current through the status LED so the LED and the Pi pin are not damaged.

- The bands look like **330 Ω** (orange-orange-black-black-brown), but the last bands can't be read in the photo.
- **Measure one with a multimeter** (the college lab has one). Any value from **220 Ω to 1 kΩ** is fine.
- If it reads about **33 Ω**, do not use it with the LED; buy two 330 Ω resistors instead.

2. Not needed now, keep for later



4 ADXL345 accelerometer module

Photo: WhatsApp Image 2026-10-15 at 1.47.06 PM

Not in the plan; works with the Pi

What it is: a small motion sensor that measures tilt, shaking and sudden falls. It connects to the Pi with 4 wires (I2C: 3.3 V, GND, SDA, SCL).

- The current plan triggers SOS with a 1-second **button press only**, so this isn't needed for the prototype.
- Possible extra later: "shake the device hard to trigger SOS" or fall detection. Keep it.
- If its header pins aren't soldered yet, the college lab can do it in a few minutes.

3. Not for this device

These parts are from temperature or energy-meter projects. They are fine parts, but our SOS device has no use for them. Don't wire them to the Pi.



5 PT100 temperature probe

Photo: 1.45.54 PM

Not needed

Steel probe on a long grey 3-wire cable. It measures temperature (e.g. of liquids or machines). An SOS device doesn't measure temperature.



6 MAX31865 board

Photo: 1.46.01 PM

Not needed

Small purple board with a green screw terminal. Its only job is reading the PT100 probe (item 5), so it goes with that probe.



7 SCT-013 current clamp (100A/50mA)

Photo: 1.46.13 PM

Not needed

Blue clip with a 3.5 mm jack. It clips around a mains wire to measure electricity use. It's for power-meter projects.



8 Thermal paste (syringe)

Photo: 1.46.23 PM

Not needed

Grey paste for fitting CPU coolers. The Pi's heatsinks are already stuck on with thermal tape, so leave them as they are.



9 Wall adapter, 5 V 1 A, round barrel plug

Photo: WhatsApp Image 2026-10-15 at 1.46.42 PM

Do not use for the Pi 4

What it is: a 5 V power adapter ("OUTPUT: 5V = 1Amp") with a round DC plug.

- The Pi 4 needs **5 V 3 A** through **USB-C**. At 1 A the Pi would slow down or keep rebooting, especially while recording video.
- The round plug doesn't fit the Pi either. Don't cut it or use plug converters.

4. What we still need (Pi 4)

Buy only this list. All prices are rough estimates in ₹; compare two or three shops.

#	Item	Qty	Notes / check at the counter	Approx. ₹
1	**Raspberry Pi camera, OV5647 5 MP** (160° wide if available)	1	Listing must say **15-pin / standard Raspberry Pi CSI**. Refuse "22-pin", "for Pi 5", "for Pi Zero", ESP32-CAM, OV2640 or OV7670. Ribbon cable comes with it	350–800 (wide: 900–1,800)
2	**Mini USB microphone** (dongle)	1	Plugs into a normal USB port. Refuse sound-sensor modules (KY-037, KY-038, MAX9814, MAX4466): the Pi can't read them	150–400
3	**microSD card**, 16 or 32 GB, Class 10 / A1	1	Known brand (SanDisk, Samsung, HP, Strontium) from a proper shop. Test with H2testw at home	250–550
4	**Push button**, 6×6 or 12×12 mm tactile	2	SOS button + spare. Must spring back (not a toggle or latching switch)	10–60
5	**LED**, 3 mm or 5 mm, red or green	2	Status light + spare	5–10
6	**Female-to-male jumper wires**	6–10	Borrow from the college lab if possible	50–100
7	**USB-C charger, 5 V 3 A**	1	Most 18 W+ phone fast chargers work; borrow one if you can. Otherwise the official Raspberry Pi 15 W USB-C supply	0 / 700–1,000
8	**USB-C cable**	1	Only if the charger doesn't have a cable attached	0 / 150–300
9	**Power bank with a USB-C output**	1	For the portable demo. A teammate's 10,000 mAh bank is plenty	0 / 1,200–2,000
10	*Only if needed:* resistors 220 Ω – 1 kΩ (330 Ω ideal)	2	Only if our resistors (item 3) measure outside this range	2–5
	Total		**Borrowing the charger, cable and power bank**	**about 800–1,500**
	Total		**Buying everything new (wide camera)**	**about 3,500–6,000**

Don't buy these

Item	Why
Heatsink	Already fitted on our Pi 4
Case	We have the official case. For the demo, leave the lid off so the camera ribbon can come out and the Pi stays cool
Breadboard	We have the MB-102
Keyboard, mouse, monitor, micro-HDMI cable	The Pi is set up over Wi-Fi from a laptop (SSH)
microSD card reader	Use the laptop slot, a borrowed reader or a phone OTG reader
Fan	Optional. Only if the Pi shows heat warnings while recording with the lid off
300 mm camera cable	Optional (₹100–200). Only if the camera needs to sit away from the board, e.g. with the case lid on

If the shop has no Pi camera

Buy a **USB webcam with a built-in microphone** (e.g. Logitech C270) and skip items 1 and 2. Tell Claude, because the recording code changes slightly (ffmpeg instead of picamera2).

5. Wiring and first-day checks

- **SOS button:** between **pin 11** (GPIO 17) and a ground pin (e.g. pin 9).
- **Status LED: pin 13** (GPIO 27) → resistor → LED long leg; LED short leg → ground (e.g. pin 14).
- **Camera:** lift the CAMERA connector's latch, push the ribbon in with the **blue strip facing the USB/Ethernet ports**, press the latch down. Always do this with the power off.
- **Microphone:** any USB port.

1. Measure a resistor (item 3) with a multimeter.
2. Test the microSD card with H2testw.
3. Flash **Raspberry Pi OS Lite (64-bit)** with Raspberry Pi Imager, setting Wi-Fi and SSH in its settings.
4. Power the Pi with the 5 V 3 A USB-C charger. If it shows an "under-voltage" warning, the charger is too weak.
5. Run the plan's Phase 1 test scripts: LED blinks, button press is seen, a 10-second video has picture and sound.
6. If a part fails, take it back to the shop within their exchange period.